

VISHAL GAURAV

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PROFESSIONAL SUMMARY

B.Tech student in Artificial Intelligence and Data Engineering with hands-on experience building end-to-end data pipelines, machine learning models, and full-stack AI applications. Skilled in Python, SQL, and cloud platforms (AWS, Azure), with exposure to ETL pipeline design, predictive modeling, natural language processing, and REST API development. Completed an internship in AI and Data Engineering and independently built multiple production-style projects spanning data engineering, systems programming, and applied machine learning.

EDUCATION

Lovely Professional University 2024 - 2028

B.Tech, Artificial Intelligence and Data Engineering

Class XII - 68% | Class X - 75%

EXPERIENCE

AI and Data Intern, Futurence Technologies 19 Jun 2025 - 26 Jul 2025

- Worked on applied Artificial Intelligence and Data Engineering tasks during a five-week internship, contributing to data pipeline development, data cleaning, and machine learning model-related deliverables using Python and SQL.

PROJECTS

CleanSweep - Automated Data Ingestion Pipeline | *Python, DuckDB, dbt-core, Pandera, Prefect, Streamlit, Docker, SQL, ETL*

- Built an end-to-end ETL data engineering pipeline simulating ingestion of daily file drops from 5 suppliers, each with real-world data quality issues (inconsistent headers, merged cells, mismatched date formats, duplicate rows).
- Designed a 3-layer data architecture (Raw to Staging to Conformed) using dbt and DuckDB, with a quarantine system that isolates bad rows with logged reasons instead of failing the pipeline.
- Implemented a schema drift detector to catch unexpected columns automatically, and built a Streamlit dashboard for observability into rows ingested, quarantined, and top error types.
- Orchestrated job scheduling and automated retries with Prefect; containerized the full environment with Docker Compose and set up CI/CD automation using GitHub Actions.

Adaptive Resource Allocation System | *Python, Machine Learning, scikit-learn, psutil, Streamlit, Tkinter, Plotly, Pandas, NumPy*

- Developed a real-time system resource monitoring and optimization tool that tracks live CPU and memory usage via psutil and predicts future CPU load using a scikit-learn Linear Regression machine learning model.
- Built adaptive allocation logic that reallocates CPU when usage exceeds 80 percent, suspends the lowest-priority process when memory exceeds 800MB, and applies priority aging to prevent process starvation.
- Shipped two parallel interfaces, a Tkinter desktop application and a Streamlit web dashboard with live Plotly data visualizations, a login system, and a process manager, sharing the same core prediction and allocation logic.
- Designed a composite system health-score formula (100 minus 0.5 times CPU percent minus 0.5 times Memory percent) to summarize system performance at a glance.

College Chatbot | *FastAPI, NLTK, scikit-learn, SQLite, HTML/CSS/JavaScript*

- Built an AI-powered chatbot that answers student queries on exam schedules, library timings, attendance rules, hostel details, and faculty contacts, using TF-IDF and Logistic Regression for intent classification.
- Served responses via a FastAPI backend against a JSON intent bank, with a browser-based chat UI and conversation logging to SQLite for analytics.

CERTIFICATIONS

- Infosys Certified Database Management System

SKILLS

Programming Languages: Python, Java, PHP, TypeScript, JavaScript, HTML, CSS

Databases: SQL, PostgreSQL, MongoDB, Firebase

Data Engineering: ETL Pipelines, Power BI, Excel, Matplotlib, NumPy, Pandas

Artificial Intelligence / Machine Learning: Machine Learning, Natural Language Processing (NLP), Deep Learning, scikit-learn

Cloud Computing: AWS, Azure, LocalStack

Tools & Platforms: Git, GitHub, VS Code, Docker, Claude Code, ChatGPT, Cowork, Manus, Antigravity